YouTube Emotion Analysis Tool - Setup & Usage Guide

Overview

This tool analyzes emotions in YouTube videos using Hume.ai's Expression Measurement API. It automatically:

1. Downloads YouTube videos/shorts
2. Submits them to Hume.ai for emotion analysis
3. Processes and displays results in multiple formats

Prerequisites

1. Get Hume.ai API Key

1. Go to https://beta.hume.ai
2. Sign up or log in
3. Navigate to Settings → Profile
4. Copy your API key

2. Install Python Requirements

```bash
# Install required packages
pip install yt-dlp hume

# Or install from requirements
pip install -r requirements.txt
```

3. Set Environment Variable

```bash
# On macOS/Linux
export HUME_API_KEY="your-api-key-here"

# On Windows (PowerShell)
$env:HUME_API_KEY = "your-api-key-here"

# Or permanently (add to ~/.bashrc or ~/.zshrc)
echo 'export HUME_API_KEY="your-api-key-here"' >> ~/.bashrc
```

Usage

Basic Usage (Simplest)

```bash
python youtube_emotion_analyzer.py "https://www.youtube.com/shorts/kpQsbOueyFI"
```

With Custom API Key

```bash
python youtube_emotion_analyzer.py "YOUR_VIDEO_URL" --api-key "your-api-key"
```

Full Options

```bash
python youtube_emotion_analyzer.py "YOUR_VIDEO_URL" \
    --api-key "your-api-key" \
    --output-dir "./results" \
    --models "face,prosody,burst,language" \
    --save-json
```

Command Line Options

Option	Description	Default
`youtube_url`	YouTube video URL (required)	-
`--api-key`	Hume.ai API key	Uses `HUME_API_KEY` env var
`--output-dir`	Directory to save video and results	`./emotion_analysis`
`--models`	Comma-separated list of models to use	`face,prosody,burst`
`--save-json`	Save detailed results to JSON	False
`--job-id`	Use existing job ID to fetch results	-

Available Models

- **face**: Analyzes facial expressions (48 dimensions)
- **prosody**: Analyzes voice tone, rhythm, timbre (48 dimensions)

- **burst**: Analyzes non-verbal sounds (laughs, sighs) (48 dimensions)
- **language**: Analyzes text sentiment and meaning (53 dimensions)

Usage Examples

Example 1: Basic Analysis of Your YouTube Short

```bash
python youtube_emotion_analyzer.py "https://www.youtube.com/shorts/kpQsbOueyFI"
```

Output:

```
[*] Downloading YouTube video from:
https://www.youtube.com/shorts/kpQsbOueyFI
[✓] Downloaded: [Video Title]
[*] Preparing emotion analysis models...
[*] Submitting job to Hume.ai API...
   Models: ['face', 'prosody', 'burst']
   Input: ./emotion_analysis/kpQsbOueyFI.mp4
[✓] Job submitted successfully!
   Job ID: job_12345abcde
[*] Waiting for analysis results (Job ID: job_12345abcde)...
   [10s] Still processing...
   [20s] Still processing...
[✓] Results received!
[*] Processing predictions...

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EMOTION ANALYSIS RESULTS
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[FACIAL EXPRESSIONS]
  joy                          75.2%
  amusement                    68.5%
  excitement                   62.1%
  surprise                     45.3%
  engagement                   38.7%

[VOICE TONE & PROSODY]
  neutral                      55.3%
  positive                     48.2%
  calm                         42.1%
  confident                    35.6%
  cheerful                     28.9%

[VOCALIZATIONS (Laughs, Sighs, etc.)]
  laugh                        82.1%
  amusement_vocalizations      71.5%
  excitement_burst             58.3%

============================================================
```

Example 2: Save Detailed Results

bash

```
python youtube_emotion_analyzer.py "https://www.youtube.com/shorts/kpQsbOueyFI"
    --save-json \
    --models "face,prosody,burst,language"
```

Creates `emotion_analysis_20240618_143022.json` with detailed results.

Example 3: Check Results from Previous Job

```bash
python youtube_emotion_analyzer.py "https://www.youtube.com/shorts/kpQsbOueyFI"
    --job-id "job_12345abcde"
```

Advanced Usage: Python API

Use the analyzer as a library in your own code:

```python
```

```python
import asyncio
from youtube_emotion_analyzer import YouTubeEmotionAnalyzer

async def analyze_video():
    # Initialize
    analyzer = YouTubeEmotionAnalyzer(
        hume_api_key="your-api-key",
        output_dir="./results"
    )

    # Download video
    video_path = analyzer.download_youtube_video(
        "https://www.youtube.com/shorts/kpQsbOueyFI"
    )

    # Analyze emotions
    job_info = await analyzer.analyze_emotions(
        video_path,
        models_to_use=['face', 'prosody', 'burst']
    )
    job_id = job_info['job_id']
    print(f"Job ID: {job_id}")

    # Wait for results
    predictions = await analyzer.wait_for_results(job_id)

    # Process results
    summary = analyzer.process_predictions(predictions)

    # Display
    analyzer.print_summary(summary)

    # Save
    analyzer.save_results(summary, "my_analysis.json")

# Run
asyncio.run(analyze_video())
```

Output Files

The tool creates the following files:

```
emotion_analysis/
├── kpQsbOueyFI.mp4                    # Downloaded video
├── emotion_analysis_20240618_143022.json  # Detailed results (if --save-
json)
└── [other downloaded videos...]
```

JSON Output Structure

```json
{
  "timestamp": "2024-06-18T14:30:22.123456",
  "face_emotions": {
    "joy": 0.752,
    "amusement": 0.685,
    "excitement": 0.621
  },
  "voice_emotions": {
    "neutral": 0.553,
    "positive": 0.482,
    "calm": 0.421
  },
  "burst_emotions": {
    "laugh": 0.821,
    "amusement_vocalizations": 0.715
  },
  "language_emotions": {},
  "overall_summary": {
    "facial_expression": {...},
    "voice_tone": {...},
    "vocalizations": {...}
  }
}
```

Understanding Results

Emotion Scores

- **Range**: 0.0 to 1.0
- **Meaning**: Higher scores = higher likelihood that humans would assign that emotion label to the expression
- **Multiple emotions**: Emotions can co-occur (e.g., joy + surprise)

Model Output Dimensions

Model	Dimensions	Measures
Face	48	Facial muscle movements (Action Units) → emotion expressions
Prosody	48	Tone, rhythm, pitch, tempo → emotional undertones
Burst	48	Non-verbal vocalizations → implicit emotions
Language	53	Word choice, tone, semantics → sentiment & meaning

Interpretation Tips

1. **Multiple high scores** = Complex emotional state
2. **Neutral + positive** = Calm but positive mood
3. **Excitement + surprise** = Unexpected positive event
4. **Laughs + amusement** = Person finding something funny

Troubleshooting

Problem: "HUME_API_KEY not found"

```bash
# Solution: Set environment variable
export HUME_API_KEY="your-api-key"

# Or use command line option
python youtube_emotion_analyzer.py "URL" --api-key "your-key"
```

Problem: "Job did not complete within 600 seconds"

YouTube videos longer than ~30 minutes may take longer. Increase the timeout or check your job ID later:

```bash
# Check results later
python youtube_emotion_analyzer.py "URL" --job-id "job_xxxxx"
```

Problem: "Error downloading video"

Some videos may be age-restricted or region-locked. Ensure:

- Video is publicly accessible
- You're not geo-blocked
- Video URL is correct

Problem: "Expression Measurement API is being sunset"

Important Note: As of June 14, 2026, the Expression Measurement API is being sunset. Check Hume.ai's current documentation for active APIs.

API Limits & Costs

- **Free tier**: Limited jobs per month
- **Processing time**: Usually 2-10 minutes per video
- **File size**: Check Hume.ai docs for limits
- **Concurrent jobs**: Maximum 500 queued jobs

Performance Tips

1. **Batch processing**: Analyze multiple videos at once
2. **Model selection**: Use only needed models (`--models "face"` is faster)
3. **Video quality**: Higher quality → more accurate detection
4. **Lighting**: Well-lit facial videos produce better results

Examples Output Interpretation

Highly Emotional Video

```
Joy:         85%
Amusement:   78%
Excitement:  72%
Surprise:    45%
→ Person is having a very positive, fun experience
```

Neutral/Skeptical Video

```
Doubt:       62%
Confusion:   58%
Concentration: 55%
```

```
Interest:   48%
→ Person is skeptical or thinking critically
```

Distressed Video

```
Sadness:    78%
Frustration: 72%
Anger:      65%
Confusion:  52%
→ Person is experiencing negative emotions
```

Additional Resources

- **Hume.ai Docs**: https://dev.hume.ai/docs/introduction
- **Semantic Space Theory**: https://www.hume.ai/publications
- **GitHub Examples**: https://github.com/HumeAI/hume-api-examples
- **API Reference**: https://dev.hume.ai/reference

Questions & Support

1. Check Hume.ai documentation
2. Review error messages carefully
3. Verify API key is correct
4. Ensure video is public/accessible

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